

Dexter Fernandes

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PLC Programmer

PLC engineer with hands-on Siemens S7-1200 / TIA Portal V13 programming experience. Built Ladder/FBD logic and tuned P/PI/PID loops for thermal processes, DC drive speed control, and sensor-driven bottling/conveyor systems. Collaborated with cross-functional, multinational teams (British Petroleum, Austin Industries) across two countries, turning technical results into clear PowerPoint decks and client-ready data visualisations. I look forward to modernise legacy BMS and plant controls cutting energy use, trimming maintenance, and easing obsolescence with retrofit-ready solutions.

WORK EXPERIENCE

Esbaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Developed time-series machine learning models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation, achieving a 23% higher F1 score than human experts.
- Led strategic MLOps initiatives to continuously enhance experimentation and deployment workflows through bayesian hyperparameter tuning of time-series and vision models.
- Spearheaded the end-to-end development and production rollout of YOLOv5 models, to elevate rig worker and equipment detection precision by 15% in 3 months.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Developed a tool for collecting raw sensor data and applying automated augmentation and labeling workflows to boost dataset diversity by 40% and cut manual annotation effort by 60%.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI • Full-time

- Led a 4-engineer team on ViCAS to design, train, and deploy deep learning models for real-time object detection, pose estimation, and tracking on edge hardware, enhancing model accuracy and reducing false positives by 33% while achieving sub-50 ms inference latency within 6 months.
- Developed and deployed core perception algorithms on resource-constrained Intel edge devices, leveraging mixed-precision quantization and structured pruning to enable real-time decision-making in harsh industrial environments.
- Implemented rigorous testing and validation frameworks based on simulated adverse conditions, to ensure model accuracy and stability, reducing performance variance by 25% under challenging lighting, occlusion, and noise scenarios.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Closed-loop oven temperature control: Implemented P/PI/PID strategies driving a 4–20 mA thyristor power unit to modulate heater output; calibrated analog I/O, added exhaust-fan load disturbances to test robustness, and tuned gains to maintain setpoint within $\pm 2^\circ\text{C}$ during transients.
- DC motor speed control via semi-controlled thyristor bridge (0.5 HP shunt motor, 24 V encoder, PID): Wrote Ladder/FBD logic to generate firing-angle commands, scaled encoder feedback, and tuned PID gains to hold setpoint speed under load changes while reducing overshoot and steady-state error.
- Automated bottling station (dispense + rotary table): Programmed sequence control for an inductive alignment sensor, capacitive level sensor, solenoid fill valve, and stepper-driven turntable.

SKILLS

- **PLC:** Siemens S7-1200, Totally Integrated Automation Portal V13
- **Software and Data:** Python
- **Business & Communication Tools:** Microsoft Powerpoint, Microsoft Excel